

3.1.4 Qualitative analysis

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Tests for ions	
(a) qualitative analysis of ions on a test-tube scale; processes and techniques needed to identify the following ions in an unknown compound:	
(i) anions:	Sequence of tests required is carbonate, sulfate then halide. (BaCO_3 and Ag_2SO_4 are both insoluble.)
CO_3^{2-}, by reaction with $\text{H}^+(\text{aq})$ forming $\text{CO}_2(\text{g})$ (see 2.1.4 c) SO_4^{2-}, by precipitation with $\text{Ba}^{2+}(\text{aq})$ Cl^-, Br^-, I^- (see 3.1.3 g)	PAG4 HSW4 Qualitative analysis.
(ii) cations: NH_4^+ , by reaction with warm $\text{NaOH}(\text{aq})$ forming NH_3 .	